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一、填空题

 1. 已知等差数列 $\{a_n\}$, $a_1 + a_3 + a_5 = 4$, $a_2 + a_4 + a_6 = 8$, 则 $a_4 + a_6 + a_8 =$ 16

 设公差为 d , 则 $a_2 + a_4 + a_6 = (a_1 + a_3 + a_5) + 3d \Rightarrow 4 + 3d = 8 \Rightarrow 3d = 4$
 $\therefore a_4 + a_6 + a_8 = (a_1 + a_3 + a_5) + 9d = 4 + 3 \times 4 = 16$

 法二: $a_1 + a_3 + a_5, a_2 + a_4 + a_6, a_3 + a_5 + a_7, a_4 + a_6 + a_8, \dots$ 是等差数列. 公差为 d

 2. 已知等差数列 $\{a_n\}$ 的前 n 项和为 S_n , 且 $a_3 + 3a_6 + a_9 = 15$, 则 S_{11} 等于 33 $\because a_4 + a_6 + a_8 = 8 + 4 \times 2$

$$a_3 + a_9 = 2a_6 \Rightarrow 5a_6 = 15 \Rightarrow a_6 = 3$$

$$\therefore S_{11} = \frac{11(a_1 + a_{11})}{2} = 11a_6 = 33$$

 3. 已知等差数列 $\{a_n\}$ 中, $a_2 = 6, a_5 = 15$, 若 $b_n = a_{2n}$, 则数列 $\{b_n\}$ 的前 5 项和等于 90

$$d = \frac{15 - 6}{3} = 3 \Rightarrow a_n = 6 + (n - 2) \cdot 3 = 3n$$

$$\therefore b_n = a_{2n} = 6n$$

$$\therefore S_5 = 5a_3 = 5 \times 18 = 90$$

 4. 已知等差数列 $\{a_n\}$ 中, $d = 2, S_{40} = 86$, 则 $a_2 + a_4 + \dots + a_{40} =$ 63

$$\begin{aligned} \text{法一: 设 } S_{\text{奇}} = a_1 + a_3 + \dots + a_{39} = A \quad \text{①} \\ S_{\text{偶}} = a_2 + a_4 + \dots + a_{40} = B \quad \text{②} \Rightarrow \begin{cases} A + B = S_{40} = 86 \\ B - A = 20d = 40 \end{cases} \Rightarrow B = 63 \end{aligned}$$

$$\text{法二: } S_{40} = 40a_1 + \frac{40 \times 39}{2} \times 2 = 86 \Rightarrow a_1 = -\frac{737}{20} \Rightarrow a_2 = -\frac{697}{20} \Rightarrow S_{\text{偶}} = -\frac{697}{20} \times 20 + \frac{20 \times 19}{2} \times 4 = 63$$

 5. 已知各项均为正数的等比数列 $\{a_n\}$ 中, 若 $a_5 a_9 = 3$, $a_6 a_{10} = 9$, 则 $a_7 a_8 =$ $3\sqrt{3}$

$$a_n > 0, q > 0$$

$$a_6 a_{10} = a_5 a_9 \cdot q^2 \Rightarrow 3q^2 = 9 \Rightarrow q = \sqrt{3}$$

$$\therefore a_7 a_8 = a_5 a_9 \cdot q = 3\sqrt{3}$$

 6. 已知首项为 1 的等比数列 $\{a_n\}$ 满足 $a_2, a_3, a_4 - 2$ 成等差数列, 则公比 $q =$ 2

$$2a_3 = a_2 + a_4 - 2 \Rightarrow 2q^2 = q + q^3 - 2$$

$$\therefore q^3 - 2q^2 + q - 2 = 0 \Rightarrow q = 2$$

 7. 已知等比数列 $\{a_n\}$ 的公比 $q = 2$, 前 n 项和为 S_n . 若 $S_3 = \frac{7}{2}$, 则 S_6 等于 $\frac{63}{2}$

$$S_3 = \frac{a_1(1 - q^3)}{1 - q} = \frac{7}{2} \Rightarrow S_6 = \frac{a_1(1 - q^6)}{1 - q} = \frac{a_1(1 - q^3)(1 + q^3)}{1 - q} = S_3 \cdot (1 + q^3) = \frac{7}{2} \times 9 = \frac{63}{2}$$

 8. 等差数列 $\{a_n\}$ 中, S_n 为其前 n 项和, 若 $\frac{S_3}{S_6} = \frac{1}{3}$, 则 $\frac{S_6}{S_{12}} =$ $\frac{3}{10}$

 设 $S_3 = t$, 则 $S_6 = 3t$, $\{a_n\}$ 是等差数列

 $\therefore S_3, S_6 - S_3, S_9 - S_6, S_{12} - S_9$ 为等差数列.

 即 $t, 2t, 3t, 4t$ 是等差数列.

$$\therefore S_{12} = t + 2t + 3t + 4t = 10t \Rightarrow \frac{S_6}{S_{12}} = \frac{3}{10}$$

t	$2t$	$3t$	$4t$
S_3	$S_6 - S_3$	$S_9 - S_6$	$S_{12} - S_9$

9. 如图, 将一个边长为 1 的正三角形的每条边三等分, 以中间一段为边向形外作正三角形, 并擦去中间一段, 得图 (2), 如此继续下去, 得图 (3) ... 试用 n 表示出第 n 个图形的边数

$$a_n = 3 \cdot 4^{n-1}$$

$$a_1 = 3, a_2 = 3 \times 4 = 12, a_3 = 12 \times 4 = 48$$

$$\frac{a_{n+1}}{a_n} = 4$$

$\therefore \{a_n\}$ 是以 3 为首项, 4 为公比的等比数列

$$\therefore a_n = 3 \cdot 4^{n-1}$$

10. 若一个等差数列前 3 项的和为 34, 最后 3 项的和为 146, 且所有项的和为 390, 则这个数列有 13 项.

$$a_1 + a_2 + a_3 = 34$$

$$a_n + a_{n-1} + a_{n-2} = 146$$

$$\therefore a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2}$$

$$\therefore 3(a_1 + a_n) = 180 \Rightarrow a_1 + a_n = 60$$

$$\therefore S_n = \frac{n(a_1 + a_n)}{2} = 30n = 390$$

$$\therefore n = 13$$

11. 已知数列 $\{a_n\}$ 的通项公式为 $a_n = (n+1)\left(\frac{10}{11}\right)^n$, ($n \in \mathbb{N}, n > 0$), 其最大项的项数是 9 或 10

$$a_{n+1} - a_n \geq 0 \Rightarrow (n+2)\left(\frac{10}{11}\right)^{n+1} - (n+1)\left(\frac{10}{11}\right)^n \geq 0 \quad \text{两边同乘 } \left(\frac{10}{11}\right)^{-(n+1)}$$

$$(n+2) - (n+1) \cdot \frac{11}{10} \geq 0 \Rightarrow n \leq 9$$

$$\therefore \textcircled{1} n=9 \text{ 时, } a_{n+1} = a_n \text{ 即 } a_9 = a_{10}$$

$$\textcircled{2} 1 \leq n \leq 9, a_{n+1} > a_n \text{ 即 } a_1 < a_2 < a_3 < \dots < a_9 < a_{10}$$

$$\textcircled{3} n \geq 10, a_{n+1} < a_n \text{ 即 } a_{10} > a_{11} > a_{12} > \dots$$

$$\therefore (a_n)_{\max} = a_9 = a_{10}$$

12. 已知各项为正数的等比数列 $\{a_n\}$ 满足 $a_7 = a_6 + 2a_5$, 若存在两项 a_m, a_n , 使得

$$\sqrt{a_m \cdot a_n} = 2\sqrt{2}a_1, \text{ 则 } \frac{1}{m} + \frac{3}{n} \text{ 的最小值为 } \underline{\frac{3}{2}}$$

$$a_n > 0, q > 0$$

$$a_5 q^2 = a_5 q + 2a_5$$

$$\therefore q^2 - q - 2 = 0 \Rightarrow q = 2 \text{ 或 } q = -1 \text{ (舍)}$$

$$\therefore \sqrt{a_m a_n} = 2\sqrt{2}a_1 \Rightarrow a_m a_n = a_1 \cdot 2^{m-1} \cdot a_1 \cdot 2^{n-1} = 8a_1^2$$

$$\therefore m+n-2=3 \Rightarrow m+n=5, m, n \in \mathbb{N}^*$$

$$\therefore \begin{cases} m=1 \\ n=4 \end{cases} \text{ 或 } \begin{cases} m=2 \\ n=3 \end{cases} \text{ 或 } \begin{cases} m=3 \\ n=2 \end{cases} \text{ 或 } \begin{cases} m=4 \\ n=1 \end{cases}$$

$$\therefore \text{当 } m=2, n=3 \text{ 时, } \left(\frac{1}{m} + \frac{3}{n}\right)_{\min} = \frac{1}{2} + 1 = \frac{3}{2}$$

二、选择题

13. 已知无穷等比数列 $\{a_n\}$ 的前 n 项和 $S_n = \frac{1}{3^n} + a (n \in N^*)$, 且 a 是常数, 则 $\sum_{i=1}^{\infty} a_i$ 是 (C)

A. $\frac{1}{3}$ B. $-\frac{1}{3}$ C. -1 D. 1

$\because \{a_n\}$ 是等比数列, $\therefore S_n = Aq^n - A \Rightarrow A = -1$

$\therefore a_1 = S_1 = \frac{1}{3} - 1 = -\frac{2}{3}, q = \frac{1}{3}$

$\therefore \sum_{i=1}^{+\infty} a_i = \lim_{n \rightarrow +\infty} S_n = \frac{a_1}{1-q} = -1$

14. 已知数列 $\{a_n\}$ 是等差数列, ($n \in N, n > 0$) 下列命题中假命题为 (D)

A. $\{a_{2n}\}$ 一定是等差数列 $\checkmark$

B. $\{a_n + a_{n+1}\}$ 一定是等差数列 $\checkmark$

C. $\{3a_n + 1\}$ 一定是等差数列 $\checkmark$

D. $\{|a_n|\}$ 一定是等差数列

A. $a_2, a_4, a_6, a_8, \dots$

$\because a_{2n} - a_{2(n-1)} = a_{2n} - a_{2n-2} = 2d$ 为定值

$\therefore \{a_{2n}\}$ 是等差数列

B. $a_1 + a_2, a_2 + a_3, a_3 + a_4, \dots$

$\because n \geq 2, (a_n + a_{n+1}) - (a_{n-1} + a_n) = (a_n - a_{n-1}) + (a_{n+1} - a_n) = 2d$ 为定值

$\therefore \{a_n + a_{n+1}\}$ 是等差数列

C. $n \geq 2, 3a_n + 1 - (3a_{n-1} + 1) = 3(a_n - a_{n-1}) = 3d$

$\therefore \{3a_n + 1\}$ 是等差数列

D. 取 $a_n = 3 - n$, 则 $\{|a_n|\}: 2, 1, 0, 1, 2, 3, \dots$

$\therefore \{|a_n|\}$ 不是等差数列

15. 等差数列 $\{a_n\}$ 的前 n 项和记为 S_n , 若已知 $a_2 + a_4 + a_{15}$ 的值, 可以求出数列 $\{S_n\}$ 中的项是 (C)

A. S_7

B. S_8

C. S_{13}

D. S_{15}

$a_2 + a_4 + a_{15} = a_2 + a_2 + 2d + a_2 + 13d = 3a_2 + 15d = 3(a_2 + 5d) = 3a_7$

$\therefore S_{13} = \frac{13(a_1 + a_{13})}{2} = 13a_7$

$\therefore$ 可求出 S_{13} 的值

拓展: $\{a_n\}$ 是等差数列, 且 $m+n+k = p+q+r = 3t, m, n, k, p, q, r, t \in N^*$

则 $a_m + a_n + a_k = a_p + a_q + a_r = 3a_t$

16. 已知项数为偶数的等比数列 $\{a_n\}$ 中, $a_1 = 1$, 奇数项和为 85, 偶数项和为 170, 则它的公

比与项数分别是 (D)

A. $-2, 4$

B. $2, 4$

C. $-2, 8$

D. $2, 8$

设: 共 n 项, 显然 $q \neq 1$

$a_1, a_3, a_5, \dots$ 是以 a_1 为首项, q^2 为公比的等比数列

$S_{\text{奇}} = \frac{a_1(1-q^n)}{1-q^2} = 85$ ①

$a_2, a_4, a_6, \dots$ 是以 a_2 为首项, q^2 为公比的等比数列

$S_{\text{偶}} = \frac{a_2(1-q^n)}{1-q^2} = 170$ ②

$\therefore \frac{\text{②}}{\text{①}}$ 得 $\frac{a_2}{a_1} = 2 = q$ 代入①

$\frac{1(1-2^n)}{1-4} = 85 \Rightarrow n = 8$

三、解答题

17. 已知数列 $\{a_n\}$ 的前 n 项和 $S_n (n \in N, n > 0)$ 满足 $2S_n = 3a_n - a_1$, 且 $a_2 + 2$ 是 a_1, a_3 的

等差中项, $\{b_n\}$ 是等差数列, $b_2 = a_2, b_8 = a_3$. (1) 求数列 $\{a_n\}, \{b_n\}$ 的通项公式;

(2) $c_n = a_n b_n (n \in N, n > 0)$, 运用推导等比数列 S_n 的方法求数列 $\{c_n\}$ 的前 n 项和 T_n .

$$\text{解: (1) } n \geq 2, 2S_{n-1} = 3a_{n-1} - a_1$$

$$\therefore n \geq 2, 2a_n = 3a_n - 3a_{n-1}$$

$$\therefore n \geq 2, a_n = 3a_{n-1} \text{ 即 } \frac{a_n}{a_{n-1}} = 3$$

$$\text{又 } 2(a_2 + 2) = a_1 + a_3 \text{ 且 } a_2 = 3a_1, a_3 = 9a_1$$

$$\therefore 2(3a_1 + 2) = a_1 + 9a_1 \Rightarrow a_1 = 1$$

$\therefore \{a_n\}$ 是等比数列.

$$\therefore a_n = 3^{n-1}$$

$$\therefore b_2 = a_2 = 3, b_8 = a_3 = 9$$

$$\therefore d = \frac{9-3}{6} = 1$$

$$\therefore b_n = 3 + (n-2) = n+1$$

$$\therefore a_n = 3^{n-1}, b_n = n+1$$

$$(2) c_n = (n+1) \cdot 3^{n-1}$$

$$T_n = 2 + 3 \cdot 3^1 + 4 \cdot 3^2 + 5 \cdot 3^3 + \dots + (n+1) \cdot 3^{n-1} \quad ①$$

$$3T_n = 2 \cdot 3^1 + 3 \cdot 3^2 + 4 \cdot 3^3 + \dots + n \cdot 3^{n-1} + (n+1) \cdot 3^n \quad ②$$

$$① - ②: -2T_n = 2 + 3^1 + 3^2 + 3^3 + \dots + 3^{n-1} - (n+1) \cdot 3^n$$

$$= 2 - (n+1) \cdot 3^n + \frac{3(1-3^{n-1})}{1-3}$$

$$= 2 - (n+1) \cdot 3^n + \frac{3(1-3^{n-1})}{-2}$$

$$\therefore T_n = -1 + \frac{n+1}{2} \cdot 3^n + \frac{3-3^n}{4} = \frac{2n+1}{4} \cdot 3^n - \frac{1}{4}$$

18. 已知等比数列 $\{a_n\}$ 中, $a_2 = 2, a_5 = 128$. 若 $b_n = \log_2 a_n$, 数列 $\{b_n\}$ 前 n 项的和为 S_n .

(1) 若 $S_n = 35$, 求 n 的值; (2) 求不等式 $S_n < 2b_n$ 的解集.

$$\text{解: (1) } a_5 = a_2 q^3 \Rightarrow q^3 = 64 \Rightarrow q = 4$$

$$\therefore a_n = 2 \cdot 4^{n-2} = 2^{2n-3}$$

$$\therefore b_n = \log_2 2^{2n-3} = 2n-3$$

$$\therefore b_{n+1} - b_n = 2 \text{ 为定值}$$

$\therefore \{b_n\}$ 是以 -1 为首项, 2 为公差的等差数列

$$\therefore S_n = -n + \frac{n(n-1)}{2} \cdot 2 = n^2 - 2n = 35$$

$$\therefore n = 7$$

$$(2) \text{ 由 (1) } S_n = n^2 - 2n, b_n = -1 + 2(n-1) = 2n-3$$

$$\therefore n^2 - 2n < 2(2n-3)$$

$$\therefore n^2 - 6n + 6 < 0$$

$$3-\sqrt{3} < n < 3+\sqrt{3} \text{ 且 } n \in N^*$$

$$\therefore n \in \{2, 3, 4\}$$

19. 设 S_n 为数列 $\{a_n\}$ 的前 n 项和, 已知 $a_n > 0$, 且 a_n, S_n, a_n^2 成等差数列.

(1) 求数列 $\{a_n\}$ 的通项公式; (2) 设 $b_n = \begin{cases} a_n, n \text{ 为奇数} \\ \frac{1}{a_n a_{n+2}}, n \text{ 为偶数} \end{cases}$, 求数列 $\{b_n\}$ 的前 20 项和 T_{20} .

解: (1) $2S_n = a_n^2 + a_n, a_n > 0$
 $n \geq 2, 2S_{n-1} = a_{n-1}^2 + a_{n-1}$

$\therefore n \geq 2, 2a_n = a_n^2 - a_{n-1}^2 + a_n - a_{n-1}$

$\therefore n \geq 2, (a_n + a_{n-1})(a_n - a_{n-1}) = a_n - a_{n-1}$

$\therefore n \geq 2, a_n - a_{n-1} = 1$

又 $2S_1 = 2a_1 = a_1^2 + a_1 \Rightarrow a_1 = 1$

$\therefore a_n = 1 + (n-1) = n$

(2) $b_n = \begin{cases} n, n \text{ 为奇数} \\ \frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2}), n \text{ 为偶数} \end{cases}$

(分析: $1, \frac{1}{2}(\frac{1}{2} - \frac{1}{4}), 3, \frac{1}{2}(\frac{1}{4} - \frac{1}{6}), 5, \frac{1}{2}(\frac{1}{6} - \frac{1}{8}) \dots$)

$\therefore S_{20} = 1 + 3 + 5 + \dots + 19 + \frac{1}{2}(\frac{1}{2} - \frac{1}{4} + \frac{1}{4} - \frac{1}{6} + \dots + \frac{1}{20} - \frac{1}{22})$
 $= \frac{10(1+19)}{2} + \frac{1}{2}(\frac{1}{2} - \frac{1}{22})$
 $= 100 + \frac{5}{22}$
 $= \frac{2205}{22}$

20. 某生产厂商为提升生产工艺, 研发了新的生产线. 新生产线的使用寿命为 4 年. 该厂商原来每个月能生产产品 1000 个, 产品合格率为 85%. 从 2021 年 1 月初开始使用新的生产线后, 每个月的产量都在前一个月的基础上提高 5%, 同时, 当月产品的不合格率与前一个月产品的不合格率之差为 -0.3% . (1) 求 2021 年 12 月的产量以及不合格品的数量;

(结果精确到 1 个); (2) 在生产过程中, 若出现某个月的不合格产品数量超过 300 个, 则生产线会被强制停用. 请分析该生产线是否存在由于上述原因而被强制停用的可能? 并说明理由.

解: (1) 设 2021 年第 n 月产量和不合格品率分别为 a_n, b_n

$\{a_n\}$ 是等比数列, $a_1 = 1000, q = \frac{a_{n+1}}{a_n} = 1.05, a_n = 1000 \cdot 1.05^{n-1}$

$\{b_n\}$ 是等差数列, $b_1 = 0.15, d = b_{n+1} - b_n = -0.003, b_n = 0.15 - 0.003(n-1) = -0.003n + 0.153$

$\therefore a_{12} = 1000 \cdot 1.05^{11} = 1710$

$b_{12} = 0.117$, 不合格数量为 $1710 \times 0.117 \approx 200$

$\therefore$ 产量为 1710 个, 不合格数量为 200 个

(2) 设第 n 月不合格产品数量为 $C_n, C_n = a_n b_n = (-0.003n + 0.153) \cdot 1000 \cdot 1.05^{n-1} = (-3n + 153) \cdot 1.05^{n-1}$

$C_{n+1} - C_n = (-3n + 150) \cdot 1.05^n - (-3n + 153) \cdot 1.05^{n-1} \geq 0$. 两边同乘 $1.05^{-(n-1)}$

$(-3n + 150) \cdot 1.05 - (-3n + 153) \geq 0 \Rightarrow -\frac{63}{20}n + \frac{315}{2} + 3n - 153 \geq 0 \Rightarrow n \leq 30$

$\therefore$ 当 $n = 30$ 时, $C_{n+1} = C_n$ 即 $C_{30} = C_{31}$. 当 $1 \leq n \leq 29$ 时, $C_{n+1} > C_n$ 即 $C_1 < C_2 < C_3 < \dots < C_{29} < C_{30}$

当 $31 \leq n \leq 47$ 时, $C_{n+1} < C_n$ 即 $C_{31} > C_{32} > \dots > C_{48}$

$\therefore (C_n)_{\max} = C_{30} = C_{31} \approx 259 < 300. \therefore$ 不可能被停用.

21. 函数 $f(x) = \frac{2x+3}{3x}$, 数列 $\{a_n\}$ 对 $n \geq 2, n \in N$ 总有 $a_n = f(\frac{1}{a_{n-1}}), a_1 = 1$;

(1) 求 $\{a_n\}$ 的通项式; (2) 设 $S_n = a_1a_2 - a_2a_3 + a_3a_4 - a_4a_5 + \cdots + (-1)^{n-1}a_na_{n+1}$, 若 $S_n \geq tn^2$ 对任意 $n \in N^*$ 恒成立, 求实数 t 的取值范围.

解: (1) $a_n = \frac{\frac{2}{a_{n-1}} + 3}{\frac{3}{a_{n-1}}} = \frac{2 + 3a_{n-1}}{3}$

$$\therefore a_n - a_{n-1} = \frac{2}{3}, a_1 = 1$$

$\therefore \{a_n\}$ 是等差数列

$$\therefore a_n = 1 + (n-1) \cdot \frac{2}{3} = \frac{2n+1}{3}$$

(2). 令 $b_n = (-1)^{n-1}a_na_{n+1}$

① 当 n 为偶数时, $a_1 - a_3 = a_3 - a_5 = \cdots = a_{n-1} - a_{n+1} = -\frac{4}{3}$

$$S_n = a_2(a_1 - a_3) + a_4(a_3 - a_5) + \cdots + a_n(a_{n-1} - a_{n+1})$$

$$= -\frac{4}{3}(a_2 + a_4 + \cdots + a_n) = -\frac{4}{3} \times \frac{\frac{n}{2}(\frac{5}{3} + \frac{2n+1}{3})}{2} = -\frac{n(2n+6)}{9}$$

② 当 n 为奇数时, $S_n = S_{n+1} - b_{n+1}$

$$= S_{n+1} - (-1)^n a_{n+1}a_{n+2} = S_{n+1} + a_{n+1}a_{n+2} = -\frac{(n+1)(2n+8)}{9} + \frac{(2n+3)(2n+5)}{9} = \frac{2n^2 + 6n + 7}{9}$$

$$\therefore S_n = \begin{cases} -\frac{n(2n+6)}{9}, & n \text{ 为偶数} \\ \frac{2n^2 + 6n + 7}{9}, & n \text{ 为奇数} \end{cases}$$

$$\therefore S_n \geq tn^2 \text{ 恒成立, 即 } \frac{S_n}{n^2} \geq t \text{ 恒成立}$$

$$\therefore \begin{cases} t \leq \left(-\frac{n(2n+6)}{9n^2}\right)_{\min} & (n \text{ 为偶数}) \\ t \leq \left(\frac{2n^2 + 6n + 7}{9n^2}\right)_{\min} & (n \text{ 为奇数}) \end{cases}$$

$$\therefore \text{只需 } t \leq \left(-\frac{n(2n+6)}{9n^2}\right)_{\min} \text{ 即可}$$

$$\therefore -\frac{2n^2 + 6n}{9n^2} = -\frac{2 + \frac{6}{n}}{9} \text{ 递增, } n \text{ 为偶数}$$

$$\therefore \left(-\frac{2n^2 + 6n}{9n^2}\right)_{\min} = -\frac{8+12}{36} = -\frac{5}{9},$$

$$\therefore t \leq -\frac{5}{9}$$

